Who benefits? Who is harmed?

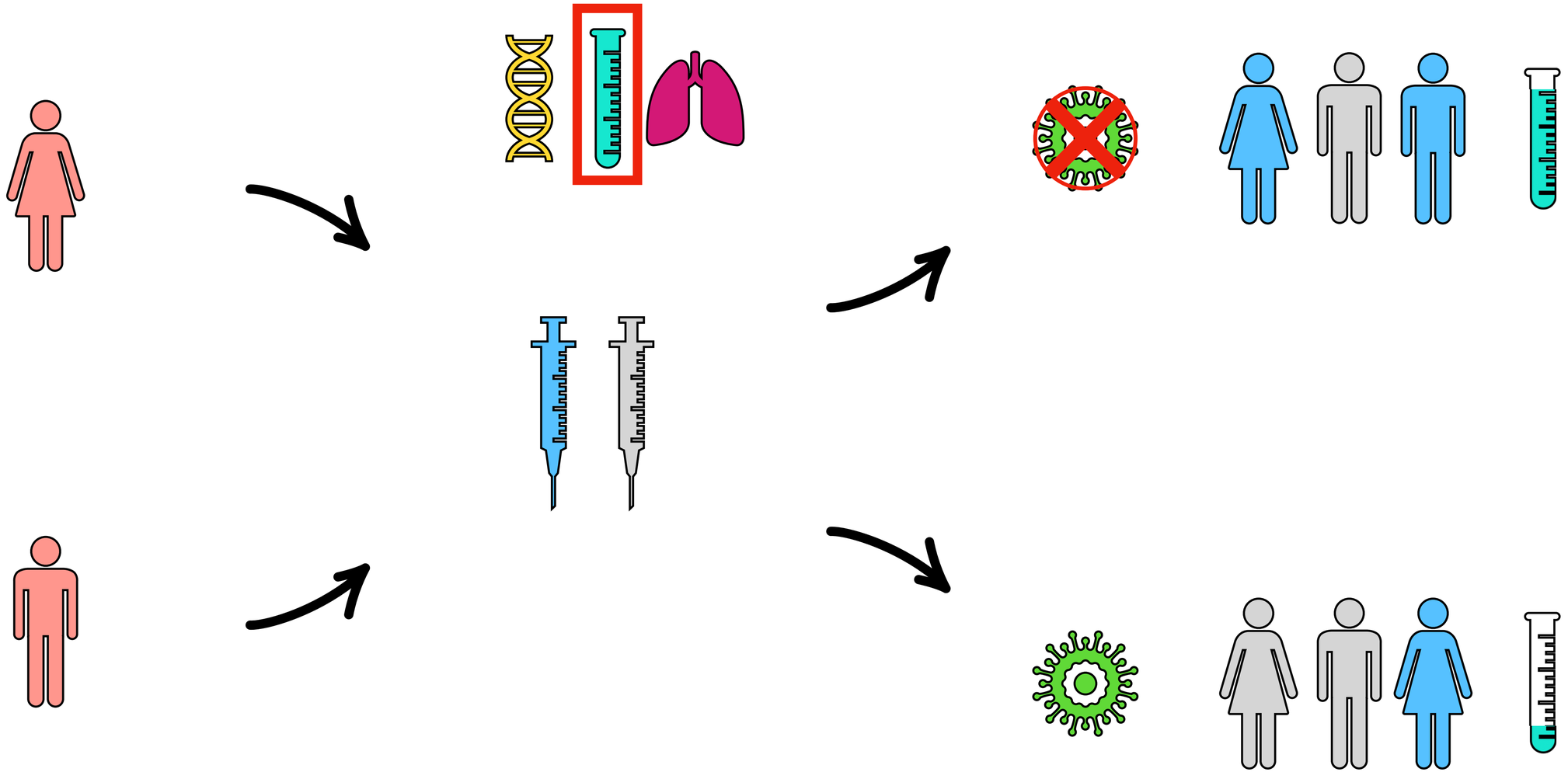
A distributional summary of causal treatment effect heterogeneity

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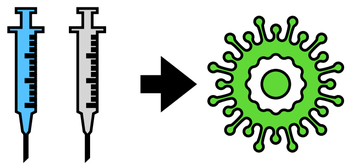
IMDS Symposium 2026

Can we identify subgroups who will respond poorly to a treatment?

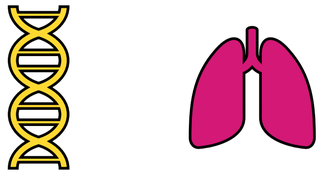


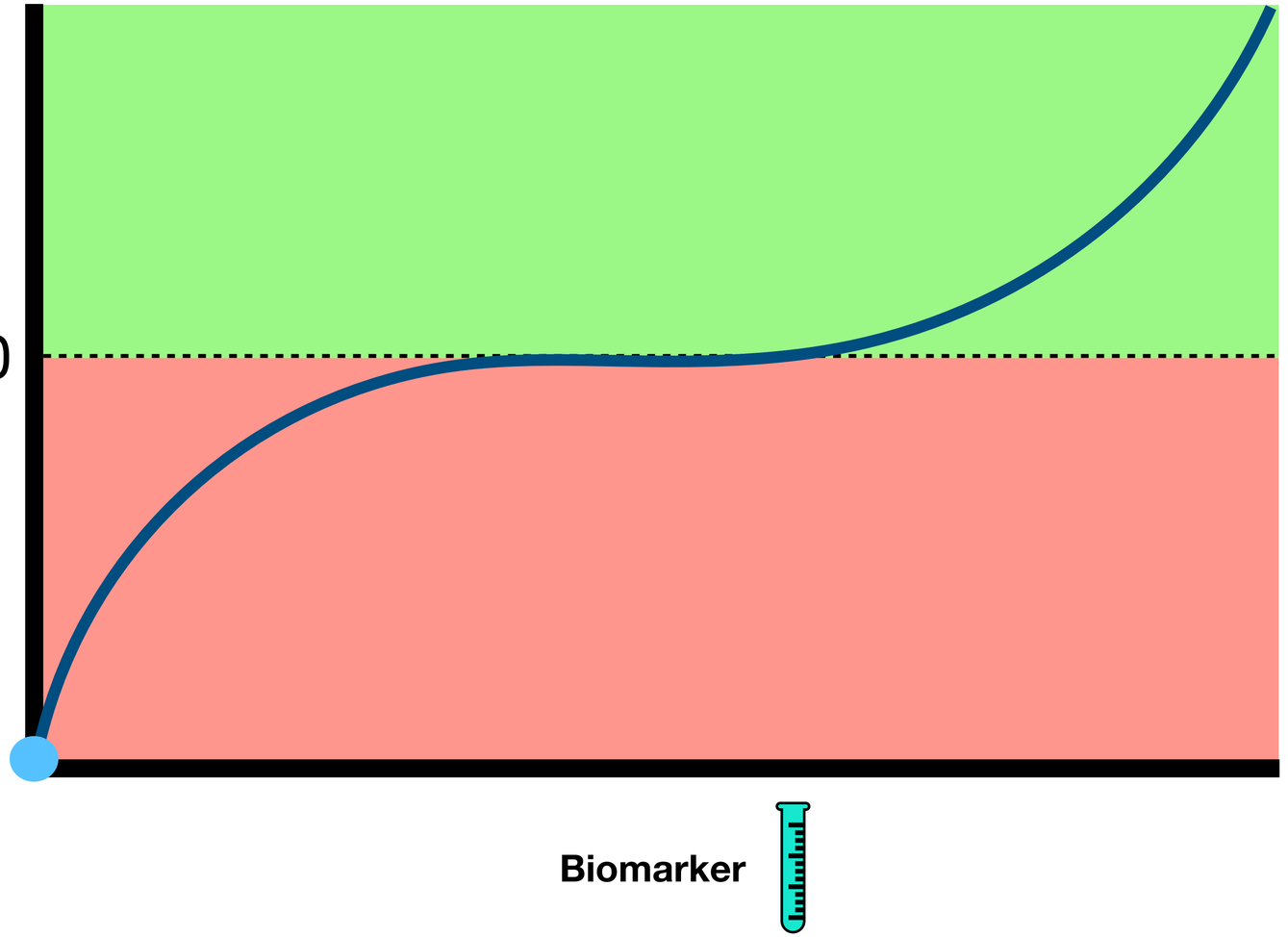
No assumption lean inference for treatment effects as a function of biomarkers

Effect of treatment on outcome

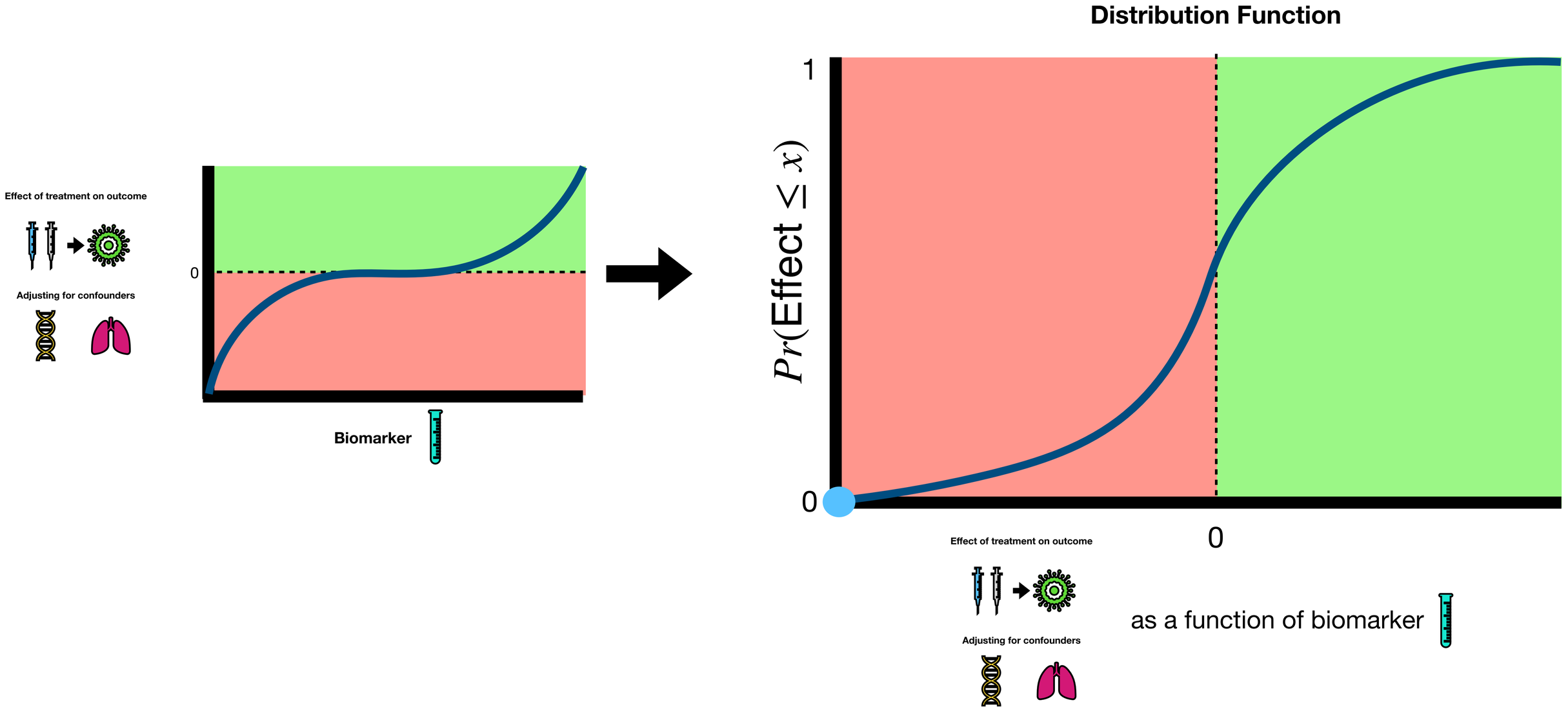


Adjusting for confounders





We obtain estimate and obtain inference for the distribution of causal effects



What do we gain by knowing the distribution of causal effects?

Smooth functionals	Average Treatment Effect, Variance,...
$\bar{\theta}_0(0)$	Fraction of individuals harmed
$1 - \bar{\theta}_0(\delta)$ for $\delta > 0$	Fraction with clinically meaningful benefit
$\bar{\theta}_0(\delta_2) - \bar{\theta}_0(\delta_1)$ for $\delta_2 > \delta_1$	Fraction with effect in $[\delta_2, \delta_1)$
$\bar{\theta}_0^{-1}(q)$ for $q \in [0,1]$	q^{th} effect quantile
$\bar{\theta}_0^{-1}(q_2) - \bar{\theta}_0^{-1}(q_2)$ for $q_2 > q_1 \in [0,1]$	quantile difference

In Summary

- Moving beyond scalar summaries is crucial for understanding treatment effect heterogeneity.

- The distribution of treatment effects provides an insightful visual summary and many useful statistics.

- We introduce a debiased machine learning + shape-constrained framework to estimate this distribution under minimal assumptions.

Thank you to my collaborators



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Thank you!

Questions?